

ONDAVELL 2mg/ml Solution for Injection

Ondansetron

Product Description

Clear and colourless solution.

The solution remain clear, colorless, and no precipitation until hour 168 (7 days) at temperature below 30°C and temperature 2-8°C when diluted in NaCl 0.9% solution, Glucose 5% solution, KCl 0.3% and NaCl 0.9% solution, KCl 0.3% and Glucose 5% solution, Mannitol 10% solution and in Ringer Solution.

Composition

Each ml solution for injection contains Ondansetron Hydrochloride Dihydrate equivalent with 2 mg of Ondansetron

Pharmacodynamics

Ondansetron is a potent, highly selective 5HT₃ receptor antagonist. Its precise mode of action in the control of nausea and vomiting is not known.

Chemotherapeutic agents and radiotherapy may cause release of 5HT in the small intestine initiating a vomiting reflex by activating vagal afferents via 5HT₃ receptors. Ondansetron blocks the initiation of this reflex.

Activation of vagal afferents may also cause a release of 5HT in the area postrema, located on the floor of the 4th ventricle, and this may promote emesis through a central mechanism. Thus, the effect of ondansetron in the management of the nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy is probably due to antagonism of 5HT₃-receptors on neurons located both in the peripheral and central nervous system. The mechanisms of actions in postoperative nausea and vomiting are not known but there may be common pathways with cytotoxic-induced nausea and vomiting. Ondansetron does not alter plasma protein concentrations.

Pharmacokinetics

Following oral administration, ondansetron is passively and completely absorbed from the gastrointestinal tract and undergoes first-pass metabolism. Peak plasma concentrations are attained approximately 1.5 hrs after dosing. For doses >8 mg, the increase in ondansetron systemic exposure with dose is greater than proportional; this may reflect reduction in first-pass metabolism at higher oral doses.

Bioavailability is slightly enhanced by the presence of food but unaffected by antacids. The disposition of ondansetron following oral, IM or IV dosing is similar with a terminal elimination half-life of about 3 hrs and steady-state volume of distribution of about 140 L. Equivalent systemic exposure is achieved after IM and IV administration of ondansetron. Ondansetron is not highly protein bound (70-76%). Ondansetron is cleared from the systemic circulation predominantly by hepatic metabolism through multiple enzymatic pathways. Less than 5% of the absorbed dose is excreted unchanged in the urine. The absence of the enzyme CYP2D6 (the debrisoquine polymorphism) has no effect on the ondansetron's pharmacokinetics. The pharmacokinetic properties of ondansetron are unchanged on repeat dosing. Studies in healthy elderly volunteers have shown slight, but clinically significant age-related increases in both oral bioavailability and half-life of ondansetron.

Gender differences were shown in the disposition of ondansetron, with females having a greater rate and extent of absorption following an oral dose and reduced systemic clearance and volume of distribution (adjusted for weight).

Indication

Management of nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy and for the prevention and treatment of postoperative nausea and vomiting.

Recommended Dose

Ondansetron is available for oral and parenteral use to allow the route of administration and dosing to be flexible.

CHEMOTHERAPY AND RADIOTHERAPY INDUCED NAUSEA AND VOMITING (CINV AND RINV)

The emetogenic potential of cancer treatment varies according to the doses and combinations of chemotherapy and radiotherapy regimens used. The selection of dose regimen should be determined by the severity of the emetogenic challenge.

Populations

CINV and RINV in Adults

Emetogenic Chemotherapy and Radiotherapy: Ondansetron can be given either by rectal, oral (tablets or syrup), intravenous or intramuscular administration.

The recommended intravenous (IV) or intramuscular (IM) dose of Ondansetron is 8 mg administered immediately before treatment.

For highly emetogenic chemotherapy, a maximum initial ondansetron dose of 16 mg IV infused over 15 minutes may be used. A single IV dose greater than 16 mg should not be given.

The efficacy of Ondansetron in highly emetogenic chemotherapy may be enhanced by the addition of a single IV dose of dexamethasone sodium phosphate, 20 mg administered prior to chemotherapy.

IV doses greater than 8 mg and up to a maximum of 16 mg must be diluted in 50 mL to 100 mL of 0.9% Sodium Chloride Injection or 5% Dextrose Injection before administration and infused over not less than 15 minutes (*see Instructions for Use and Handling*). Ondansetron doses of 8 mg or less do not need to be diluted and may be administered as a slow IM or IV injection in not less than 30 seconds.

The initial dose of Ondansetron may be followed by 2 additional IV or IM doses of 8 mg 2 to 4 hours apart, or by a constant infusion of 1 mg/h for up to 24 hours.

Oral treatment is recommended to protect against delayed or prolonged emesis after the first 24 hours.

CINV in Children and Adolescents (aged 3 years to 17 years)

The dose for CINV can be calculated based on body surface area (BSA) or weight.

Ondansetron was given by IV infusion diluted in 25 to 50 mL of saline or other compatible infusion fluid and infused over not less than 15 minutes.

Dosing by BSA

Ondansetron should be administered immediately before chemotherapy as a single IV dose of 5mg/m². The IV dose must not exceed 8 mg. Oral dosing can commence 12 hours later and may be continued for up to 5 days (Table 1). Adult doses must not exceeded.

Table 1. BSA-based dosing for CINV

BSA	Day 1	Days 2 - 6
≥ 0.6 m ² to ≤ 1.2 m ²	5 mg/m ² IV plus 4 mg tablet after 12 hours	4 mg tablet every 12 hours
> 1.2 m ²	5 mg/m ² IV or 8 mg IV plus 8 mg tablet after 12 hours	8 mg tablet every 12 hours

Dosing by body weight

Ondansetron should be administered immediately before chemotherapy as a single IV dose of 0.15 mg/kg. The IV dose must not exceed 8 mg. On Day 1, two further IV doses may be given in 4-hourly intervals. Oral dosing can commence 12 hours later and may be continued for up to 5 days (Table 2). Adult doses must not exceeded.

Table 2. Weight-based dosing for CINV

Body Weight	Day 1	Days 2 - 6
> 10 kg	Up to 3 doses of 0.15 mg/kg IV every 4 hours	4 mg tablet every 12 hours

There are no dosing recommendations for children with BSA <0.6 m², body weight ≤10 kg or who are unable to swallow tablets.

CINV and RINV in Elderly

Ondansetron is well tolerated by patients over 65 years of age.

In patients 65 years of age or older, all IV doses should be diluted and infused over 15 minutes and, if repeated, given no less than 4 hours apart.

In patients 65 to 74 years of age, the initial IV dose of Ondansetron 8 mg or 16 mg, infused over 15 minutes, may be followed by 2 doses of 8 mg infused over 15 minutes and given no less than 4 hours apart.

In patients 75 years of age or older, the initial IV dose of Ondansetron should not exceed 8 mg infused over 15 minutes. The initial dose of 8 mg may be followed by 2 doses of 8 mg, infused over 15 minutes and given no less than 4 hours apart (*see Special Patient Populations, Elderly*).

Renal Impairment

No alteration of daily dosage or frequency of dosing, or route of administration are required.

Hepatic Impairment

Clearance of ondansetron is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8 mg IV or oral should not be exceeded.

Patients with Poor Sparteine/Debrisoquine Metabolism

The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing is required.

POST-OPERATIVE NAUSEA AND VOMITING (PONV)

PONV in Adults

For the prevention of PONV: Ondansetron can be administered orally or by intravenous or intramuscular injection.

The recommended dose of Ondansetron injection is a single dose of 4 mg by IM or slow IV injection administered at the induction of anaesthesia.

For treatment of established post-operative nausea and vomiting, a single dose of 4 mg given by IM or slow IV injection is recommended.

PONV in Children and Adolescents (aged 3 years to 17 years)

For prevention and treatment of PONV in paediatric patients having surgery performed under general anaesthesia, Ondansetron may be administered by slow IV injection (not less than 30 seconds) at a dose of 0.1 mg/kg up to a maximum of 4 mg either prior to, at or after induction of anaesthesia, or after surgery.

There are no data on the use of ondansetron in the treatment of PONV in children below 2 years of age.

Elderly

There is limited experience in the use of Ondansetron in the prevention and treatment of post-operative nausea and vomiting in the elderly, however Ondansetron is well tolerated in patients over 65 years receiving chemotherapy.

Renal Impairment

No alteration of daily dosage or frequency of dosing, or route of administration are required.

Hepatic Impairment

Clearance of ondansetron is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8 mg IV or oral should not be exceeded.

Patients with Poor Sparteine/Debrisoquine Metabolism

The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently in such patients repeat dosing will give drug exposure levels no

different from those of the general population. No alteration of daily dosage or frequency of dosing is required.

Compatibility:

Ondansetron Injection is compatible towards 5% Glucose Solution; 5% Glucose Solution in Water with 0.3% Potassium Chloride Solution; 10% Mannitol Solution; Ringer Solution; 0.9% Sodium Chloride Solution; 0.9% Sodium Chloride Solution in Water with 0.3% Potassium Chloride Solution.

Incompatibilities

Ondansetron injection should not be administered in the same syringe or infusion as any other medication. Ondansetron injection should only be admixed with those infusion solutions which are recommended.

Route of Administration

Intravenous and intramuscular

Contraindications

Hypersensitivity to any component of Ondavell.

Based on reports of profound hypotension and loss of consciousness when ondansetron was administered with apomorphine hydrochloride, concomitant use with apomorphine is contraindicated.

Warnings and Precautions

Hypersensitivity reactions have been reported in patients who have exhibited hypersensitivity to other selective 5HT₃-receptor antagonists.

Very rarely and predominantly with IV ondansetron, transient electrocardiogram (ECG) changes including QT interval prolongation have been reported.

As ondansetron is known to increase large bowel transit time, patients with signs of subacute intestinal obstruction should be monitored following administration.

Interactions with other Medicaments

There is no evidence that ondansetron either induces or inhibits the metabolism of other drugs commonly co-administered with it. Specific studies have shown that ondansetron does not interact with alcohol, temazepam, frusemide, tramadol and propofol.

Ondansetron is metabolised by multiple hepatic cytochrome P-450 enzymes: CYP3A4, CYP2D6 and CYP1A2. Due to the multiplicity of metabolic enzymes capable of metabolising ondansetron, enzyme inhibition or reduced activity of 1 enzyme (eg, CYP2D6 genetic deficiency) is normally compensated by other enzymes and should result in little or no significant change in overall ondansetron clearance or dose requirement.

Phenytoin, Carbamazepine and Rifampicin: In patients treated with potent inducers of CYP3A4 (ie, phenytoin, carbamazepine and rifampicin), the oral clearance of ondansetron was increased and ondansetron blood concentrations were decreased.

Tramadol: Data from small studies indicate that ondansetron may reduce the analgesic effect of tramadol.

Pregnancy and Lactation

Pregnancy

The use of ondansetron in pregnancy is not recommended.

In human epidemiological studies, an increase in orofacial clefts was observed in infants of women administered ondansetron during the first trimester of pregnancy. Regarding cardiac malformations, the epidemiological studies showed conflicting results.

Three epidemiological studies in the US assessed the risk of specific congenital anomalies, including orofacial clefts and cardiac malformations in offspring born to mothers exposed to ondansetron during the first trimester of pregnancy.

- One cohort study with 88,467 pregnancies exposed to ondansetron showed an increased risk of oral clefts (3 additional cases per 10,000 women treated, adjusted relative risk (RR), 1.24 (95% CI 1.03-1.48)) without an apparent increase in risk of cardiac malformations. A separately published subgroup analysis of 23,877 pregnancies exposed to intravenous ondansetron did not find an increased risk of either oral clefts or cardiac malformations.
- One case-control study using population-based birth defect registries with 23,200 cases across two datasets showed an increased risk of cleft palate in one dataset and no increased risk in the other dataset. There was no increased risk of cardiac malformations in this study.
- The second cohort study with 3,733 pregnancies exposed to ondansetron found an increased risk of ventricular septal defect, adjusted RR 1.7 (95%CI 1.0-2.9), but no statistically significant increase in risk of cardiac malformations.

Reproductive studies in rats and rabbits did not show evidence of harm to the fetus.

Pregnancy status should be verified for females of reproductive potential prior to starting the treatment with Ondavell.

Females of reproductive potential should be advised that it is possible that Ondavell can cause harm to the developing fetus. Sexually active females of reproductive potential are recommended to use effective contraception (methods that result in less than 1 % pregnancy rates) when using Ondavell during the treatment and for two days after stopping treatment with Ondavell.

Side Effects

The following side effects can occur: Headache, a sensation of warmth or flushing, hiccups and occasional, asymptomatic increases in liver function tests have been reported. Ondansetron is known to increase large bowel transit time and may cause constipation in some patients.

There have been rare reports of immediate hypersensitivity reactions sometimes severe including anaphylaxis.

There have been rare reports suggestive of extrapyramidal reactions eg, oculogyric crisis/dystonic reactions without definitive evidence of persistent clinical sequelae.

Seizures have been rarely observed. There have been rare reports of chest pain with or without ST segment depression, arrhythmias, hypotension and bradycardia.

Symptoms and Treatment of Overdose

Symptoms: There is no limited experience of ondansetron overdose. In the majority of cases, symptoms were similar to those already reported in patients receiving recommended doses

Treatments: There is no specific antidote for ondansetron, therefore in cases of suspected overdose, symptomatic and supportive therapy should be given as appropriate. The use of ipecacuanha to treat overdose with ondansetron is not recommended as patients are unlikely to respond due to the antiemetic action of ondansetron itself.

Effects on the Ability to Drive or Use Machine

In psychomotor testing, ondansetron does not impair performance nor cause sedation.

Storage Conditions

Store below 30°C. Protect from light.

Ondavell Injection after being reconstitution with infusion solutions (Glucose 5%, Sodium Chloride 0.9%, Ringers Solution, Mannitol 10%, Potassium Chloride 0.3% & Sodium Chloride 0.9% Infusion, and Potassium Chloride 0.3% & Glucose Infusion 5% Infusion) remains stable for 168 hours (7 days) with storage condition at 2–8°C and at temperature below 30°C.

Presentation

ONDAVELL 2mg/ml Solution for Injection – Box of 5 ampoules @ 2 ml
– Box of 5 ampoules @ 4 ml

Shelf Life

36 months from the date of manufacture if kept as recommended.

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Registration Number

ONDAVELL 2mg/ml Solution for Injection – MAL19041006AZ

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Date of Revision of Package Insert

22nd April 2021